

Graph Inductive Biases in Transformers: Architectural Explorations and Structural Enhancements

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Abstract

Graph Neural Networks (GNNs) have traditionally relied on local message-passing paradigms, which possess strong structural biases but suffer from over-smoothing and limited long-range reasoning. Graph Transformers address long-range dependencies via global attention but are inherently structure-blind. This report explores and extends the Graph Inductive bias Transformer (GRIT), an architecture that injects structural geometry into global attention without relying on explicit message-passing modules. We specifically investigate Relative Random Walk Probabilities (RRWP) and dynamic edge updates. Furthermore, we propose and evaluate three architectural enhancements: Motif-Aware Diffusion Encoding, Pruned- K RRWP, and Adaptive- K RRWP. We benchmark these modifications on the ZINC molecular property regression dataset. Finally, we conduct additional exploratory experiments to empirically validate theoretical claims regarding shortest-path encoding and to analyze the inherent relational dynamics of the learned attention distributions.

1 Introduction

Learning effective representations for graph-structured data requires balancing local structural awareness with global context. Message-Passing Neural Networks (MPNNs) excel at capturing immediate neighborhood topology, imposing a strong local inductive bias. However, this iterative aggregation mechanism is fundamentally bottlenecked by issues such as over-smoothing and over-squashing, which restrict the network’s capacity to perform complex, long-range structural reasoning.

In contrast, Transformers employ a global attention mechanism capable of modeling long-range dependencies. The primary limitation of standard Transformers in the graph domain is their structural blindness; without explicit positional encodings, they treat graphs as unordered sets of nodes, discarding the underlying topology.

To bridge this gap, the Graph Inductive bias Transformer (GRIT) integrates the structural power of message-passing into a pure Transformer architecture. In this project, we study the core mechanisms of GRIT, most notably its use of Relative Random Walk Probabilities (RRWP) and dynamic node-pair representations. Building upon this foundation,

we propose novel structural enhancements tailored for complex chemical graphs and evaluate their efficacy on the ZINC dataset, focusing on the critical task of molecular property regression (predicting $\log P$, a key metric for aqueous solubility in drug discovery).

2 Background on GRIT

2.1 Relative Random Walk Probabilities (RRWP)

Standard transformers often rely on Shortest Path Distances (SPD) to encode graph topology. GRIT replaces simple scalar distances with RRWP, capturing a richer, higher-order statistical view of connectivity. For a graph with adjacency matrix A and degree matrix D , the random walk transition matrix is defined as $M = D^{-1}A$. The RRWP tensor explicitly encodes the probability of a walk from node i to node j in exactly k steps. The initial edge representation is initialized as a K -dimensional vector:

$$P_{i,j} = [I, M, M^2, \dots, M^{K-1}]_{i,j} \in \mathbb{R}^K \quad (1)$$

where K acts as the maximum considered walk length, serving as a receptive field hyperparameter.

2.2 Dynamic Edge Representation Updates

Unlike standard graph transformers that compute static structural biases, GRIT allows edge representations to evolve continuously at every layer. Initialized as $e_{i,j}^{(0)} = P_{i,j}$, the representations update via a Multi-Layer Perceptron (MLP) that fuses the structural edge vector with the representations of its incident nodes:

$$\hat{e}_{i,j}^{(l)} = \text{MLP} \left((W_Q^{(l)} x_i^{(l-1)} + W_K^{(l)} x_j^{(l-1)}) \odot (W_E^{(l)} e_{i,j}^{(l-1)}) \right) \quad (2)$$

where W_Q, W_K, W_E are trainable weight matrices, and $\odot$ denotes element-wise multiplication.

2.3 Structure-Aware Node Aggregation

The updated edge representations directly inform the global attention mechanism. Node representations are aggregated

not just by matching keys and queries, but by incorporating the evolved structural connections between pairs:

$$x_i^{(l)} = \sum_{j \in \mathcal{V}} \alpha_{i,j}^{(l)} \cdot \left(W_V^{(l)} x_j^{(l-1)} + W_{Ev}^{(l)} \hat{e}_{i,j}^{(l)} \right) \quad (3)$$

where the attention scores are modulated by the edge representations:

$$\alpha_{i,j}^{(l)} = \text{Softmax}_j \left(W_A^{(l)} \hat{e}_{i,j}^{(l)} \right) \quad (4)$$

This ensures that structural geometry is strictly preserved during the attention computation.

2.4 Anti-Over-Smoothing Design

Standard Transformers rely heavily on Layer Normalization, which fundamentally suppresses node degree information by normalizing features across the hidden dimension, mathematically erasing the distinction between high-degree hubs and isolated leaf nodes. GRIT circumvents this by substituting LayerNorm with Batch Normalization and explicitly injecting degree information back into the node representation prior to the feed-forward network. This design choice effectively delays over-smoothing and enables the training of deeper architectures.

3 Proposed Improvements

3.1 Motif-Aware Diffusion Encoding

While standard RRWP effectively captures global connectivity and random walk reachability, it is fundamentally an isotropic diffusion process. In domains such as molecular chemistry, chemical properties (like those targeted in the ZINC dataset) are heavily dictated by specific functional groups and cyclic substructures (e.g., aromatic carbon rings).

To transition the network from basic node-level reasoning to functional-group reasoning, we augment the edge representations with motif-level structural signals. We construct a triangle motif adjacency matrix to explicitly capture 3-cycle participation:

$$A_{\text{motif}} = (A^2) \odot A \quad (5)$$

Here, $(A^2)_{i,j}$ counts the 2-hop paths between nodes, and element-wise multiplication with A ensures the nodes are directly connected, forming a triangle. From this, we derive a motif transition matrix $M_{\text{motif}} = D_{\text{motif}}^{-1} A_{\text{motif}}$.

To allow the network to reason over both global reachability and localized ring structures, we construct a dual-channel random walk encoding. The unified edge initialization is concatenated as:

$$e_{i,j}^{(0)} = [P_{i,j}^{(1)}, M_{i,j}^{(1)}, \dots, P_{i,j}^{(K)}, M_{i,j}^{(K)}] \in \mathbb{R}^{2K} \quad (6)$$

Nodes participating in cyclic structures receive stronger structural signals, explicitly highlighting the presence of chemically relevant rings.

3.2 Pruned- K and Adaptive- K RRWP

A distinct limitation of the original GRIT framework is the reliance on a fixed maximum walk length K . Applying a “one-size-fits-all” K introduces failure modes: small graphs suffer from over-smoothed walks as the random walk rapidly hits its stationary distribution, while very large graphs may experience insufficient structural coverage. Extra walk dimensions beyond a graph’s structural saturation point simply introduce redundant noise.

We propose two methodologies to address scale variance: Pruned- K and Adaptive- K .

Pruned- K RRWP: This is a graph-aware approach that introduces a hard cutoff. We monitor the Frobenius norm difference between successive diffusion steps. If the diffusion saturates, we truncate the walk and pad the remaining dimensions with zeros:

$$\|P^k - P^{k-1}\|_F < \epsilon \quad (7)$$

This prevents the encoding from converging to a stationary distribution, removing redundant structural noise on smaller graphs.

Adaptive- K RRWP: This is a task-aware approach utilizing soft weighting. We introduce a learnable gating mechanism to emphasize task-relevant structural scales:

$$\tilde{P}_{i,j}^{(k)} = \sigma(g_k) \cdot P_{i,j}^{(k)} \quad (8)$$

By scaling the RRWP tensor, the network learns to autonomously suppress noisy diffusion scales and amplify the walk lengths that correspond to the most critical substructures for the specific regression task.

4 Experiments on ZINC

We benchmarked the proposed modifications against the baseline GRIT architecture using the ZINC molecular dataset to predict the log P regression target.

4.1 Motif-Aware Diffusion Results

Figures 1, 2, and 3 illustrate the training, validation, and test performance, respectively, comparing the original GRIT with our Motif-GRIT over 50 epochs. Qualitatively, incorporating the dual-channel motif encoding allowed the model to leverage ring-structure information early in the training process. This explicit geometric hinting generally yields more stable convergence properties, as the network does not have to implicitly learn to detect aromatic rings solely from isotropic diffusion statistics.

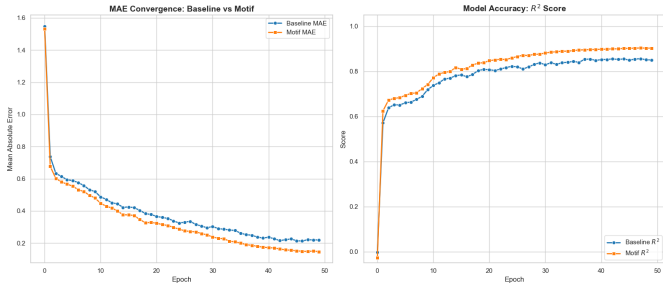


Figure 1: Training MAE and R^2 comparison between Original GRIT and Motif-GRIT over 50 epochs.

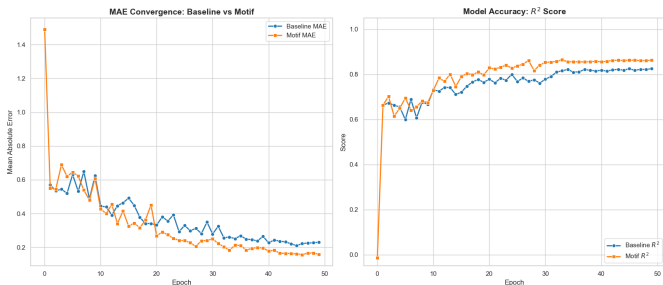


Figure 2: Validation MAE and R^2 comparison between Original GRIT and Motif-GRIT.

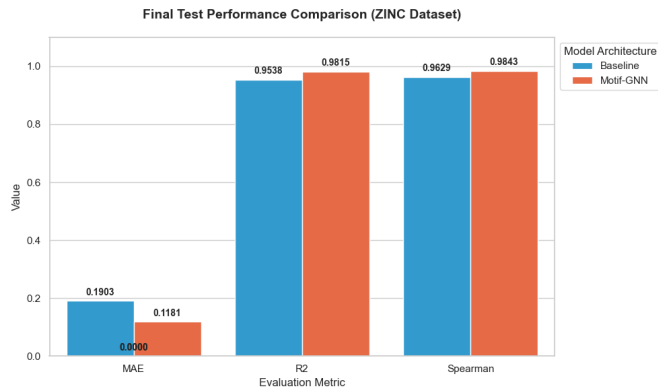


Figure 3: Test metrics comparison after 50 epochs.

4.2 Scale-Adjusted Diffusion Results

Figures 4, 5, and 6 detail the comparative performance of the baseline against the Pruned- K and Adaptive- K strategies. The hard cutoff in Pruned- K effectively mitigates over-smoothing on the smaller molecules typical of the ZINC dataset, removing redundant feature dimensions. Meanwhile, the Adaptive- K mechanism demonstrates the efficacy of letting the gradient dictate the relevance of specific diffusion scales, acting as a structural attention mechanism over the random walk steps.

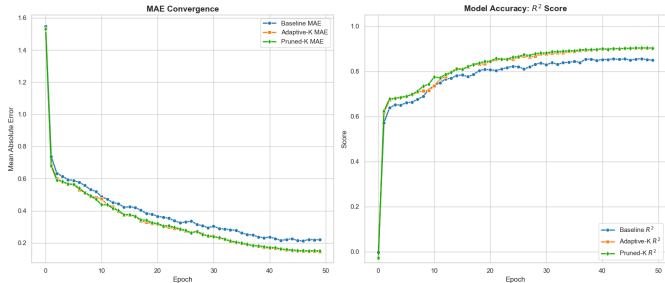


Figure 4: Training MAE and R^2 comparison among Original GRIT, Pruned- K , and Adaptive- K .

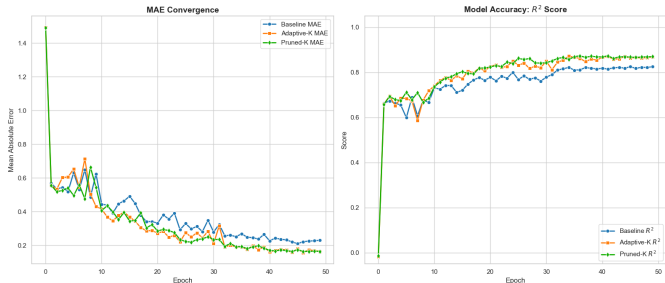


Figure 5: Validation MAE and R^2 comparison among Original GRIT, Pruned- K , and Adaptive- K .

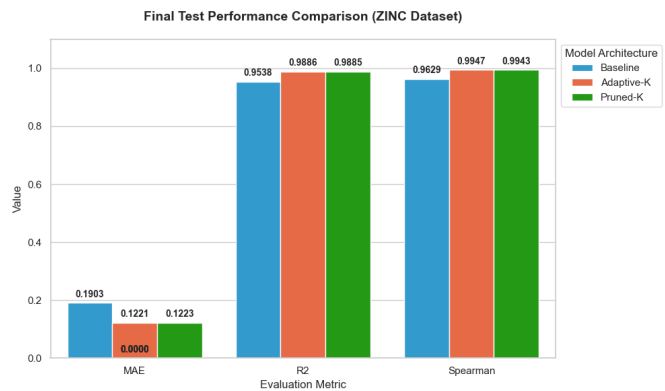


Figure 6: Test metrics comparison after 50 epochs.

5 Additional Experiments

5.1 Validating Proposition 3.1 from Paper on SPD

To rigorously understand the representational capacity of the RRWP formulation, we empirically investigated Proposition 3.1 from the original GRIT paper, which claims that an MLP operating over RRWP encodings can approximate Shortest Path Distances (SPD). This is a profound theoretical claim, as it implies that explicit graph geometry naturally emerges from purely statistical diffusion vectors.

We constructed node-pair RRWP feature vectors $[I, M, \dots, M^{K-1}]$ over synthetic graph families and trained

a lightweight multilayer perceptron (utilizing GELU activations and cross-entropy loss) to predict the truncated SPD between node pairs. The experiment yielded highly positive results: the MLP successfully and robustly recovered the exact shortest-path bounds. This validates that the K -dimensional diffusion statistics implicitly encapsulate rigid geometric distance, confirming why RRWP serves as an exceptionally strong initialization for graph transformers.

Please refer to the attached notebook `proposition.3.1.rrwp.to.spd.ipynb` on GitHub; all experimental details are documented in the markdown cells.

5.2 Attention Structure Analysis

In another exploratory experiment, we analyzed the relational structure of the learned attention distributions in a simplified GRIT-style transformer. The original motivation was to explore whether some form of structured sparsification using LSH could make sense inside GRIT. The intuition was that if most of the meaningful attention naturally concentrated around certain categories of node pairs, then it might be possible to preserve only those interactions instead of maintaining full all-pairs attention.

We introduced two categories:

- **PA (Physically Adjacent):** directly connected node pairs.
- **SSR (Structurally Similar Relations):** node pairs with highly similar RRWP fingerprints, measured using cosine similarity.

The hypothesis was that even though GRIT uses global attention, a significant fraction of the useful attention mass might naturally concentrate around PA or SSR pairs. However, the experiment did not strongly validate this idea. The learned attention distributions did not consistently cluster around these categories in a significant way.

This turned out to be informative in itself. One important realization was that such assumptions may only hold for certain task settings. Attention patterns in graph transformers appear to be highly task-dependent, and RRWP cosine similarity alone may not directly correspond to the interactions useful for downstream prediction. Additionally, since this experiment used a simplified GRIT-style architecture rather than the complete GRIT pipeline, some of the richer relational dynamics may not have been captured properly.

Even though the experiment was somewhat inconclusive, it helped us better understand that graph transformer attention is significantly more nuanced than simple notions of local adjacency or structural similarity.

6 Discussion and Limitations

While GRIT and our proposed extensions significantly improve structural reasoning, they introduce notable computational trade-offs. The dynamic updating of edge representa-

tions $e_{i,j}^{(l)}$ at every transformer layer incurs an $\mathcal{O}(N^2)$ memory footprint. While feasible for small molecular graphs like those in ZINC, this quadratic scaling poses a severe limitation for large-scale networks. Furthermore, while Motif-GRIT improves local reasoning, the explicit construction of motif transition matrices introduces preprocessing overhead that scales poorly with motif complexity.

7 Conclusion

In this project, we explored the mechanisms by which GRIT integrates inductive biases into graph transformers without explicit message passing. By analyzing the Relative Random Walk Probabilities and dynamic edge representations, we identified avenues for architectural improvement. Our implementations of Motif-Aware Diffusion, Pruned- K , and Adaptive- K RRWP successfully demonstrate that augmenting statistical diffusion with explicit geometric hinting and scale-awareness improves molecular property regression. Furthermore, our exploratory analysis confirmed the geometric expressivity of RRWP while revealing the complex, task-driven nature of learned global attention, highlighting critical areas for future scaling and structural research in Graph Machine Learning.

References

- [1] Ma, L., Lin, C., Lim, D., Romero-Soriano, A., Dokania, P. K., Coates, M., Torr, P. H. S., & Lim, S.-N. (2023). *Graph Inductive Biases in Transformers without Message Passing*. Proceedings of the 40th International Conference on Machine Learning (ICML).
- [2] Irwin, J. J., Sterling, T., Mysinger, M. M., Bolstad, E. S., & Coleman, R. G. (2012). *ZINC: A Free Tool to Discover Chemistry for Biology*. Journal of Chemical Information and Modeling.